

Impact of Remote Work on Mental Health

Analysing Mental Well-being in a Remote Work Era

Carolina Prata 114246
João Varela 113780



Phenomenon

Remote work is changing how employees handle stress, work-life balance, and mental health.

Focus areas:

- ❑ Work Environment
- ❑ Mental Health Impact
- ❑ Work-Life Balance & Satisfaction
- ❑ Demographic Patterns
- ❑ Organizational Influences



Dataset Overview

The dataset *“Impact of Remote Work on Mental Health”* from Kaggle contains **5,000** records collected from employees worldwide.

Quantitative

Age

Years_of_Experience

Hours_Worked_Per_Week

Number_of_Virtual_Meetings

Work_Life_Balance_Rating

Social_Isolation_Rating

Company_Support_for_Remote_Work

Qualitative

Stress_Level

Gender

Job_Role

Industry

Work_Location

Mental_Health_Condition

Access_to_Mental_Health_Resources

Productivity_Change

Satisfaction_with_Remote_Work

Physical_Activity

Sleep_Quality

Region

Data example

Age	Gender	Job_Role	Industry	Years_of_Experience	Work_Location	Hours_Worked_Per_Week
32	Non-binary	HR	Healthcare	13	Hybrid	47
40	Female	Data Scientist	IT	3	Remote	52

Number_of_Virtual_Meetings	Work_Life_Balance_Rating	Stress_Level	Mental_Health_Condition	Access_to_Mental_Health_Resources	Productivity_Change
7	2	Medium	Depression	No	Decrease
4	1	Medium	Anxiety	No	Increase

Social_Isolation_Rating	Satisfaction_with_Remote_Work	Company_Support_for_Remote_Work	Physical_Activity	Sleep_Quality	Region
1	Unsatisfied	1	Weekly	Good	Europe
3	Satisfied	2	Weekly	Good	Asia

Users and Context of Use



Researchers

WHY: Study how remote, hybrid, and onsite work affect mental health, and productivity

HOW: Using the provided CSV dataset and the interactive app

Main questions to answer

How do **weekly hours** and **meeting load** differ across remote/hybrid vs on-site staff?

Which regions or industries report higher **stress** or **mental-health conditions**?



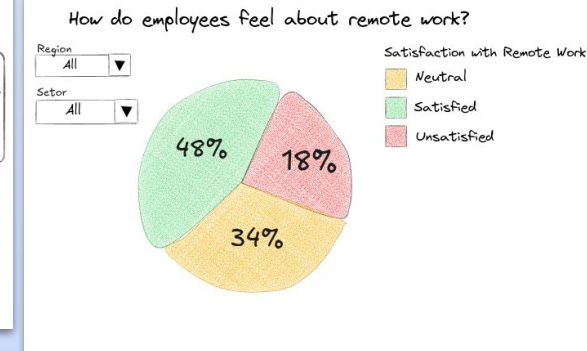
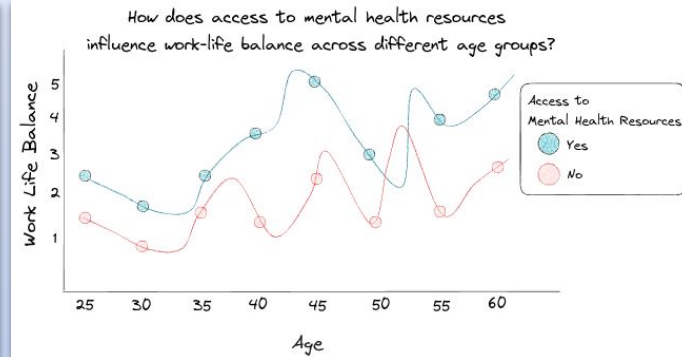
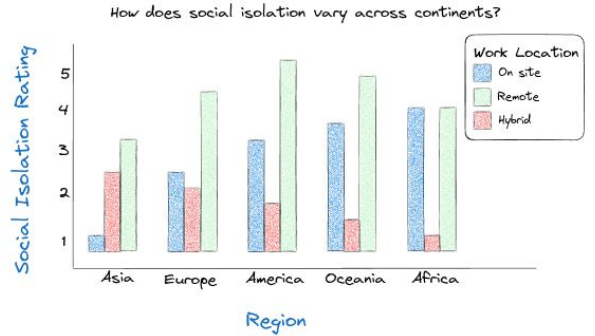
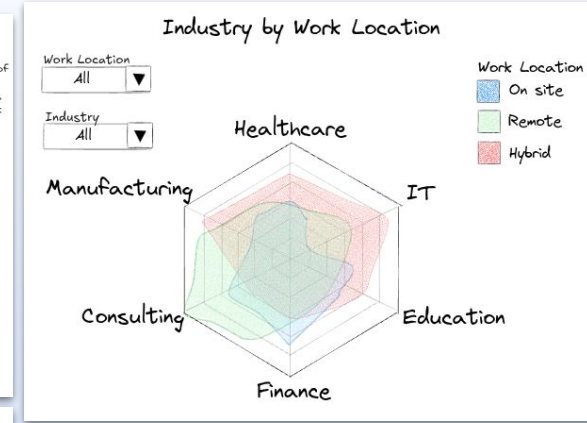
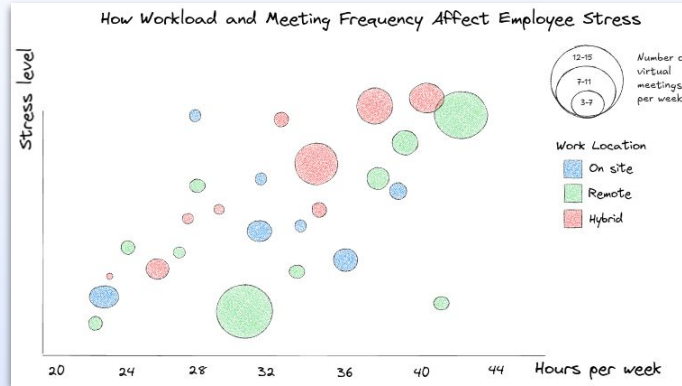
Does having **access to mental health** resources connect with better work life balance or lower social isolation ratings?

How do **sleep quality** and **physical activity** shift alongside stress levels for different work locations and genders?

User Evaluation – Paper Prototype

5 types of Visualization Techniques

- Bubble Chart
- Spider Chart
- Grouped Bar Chart
- Line Chart
- Pie Chart

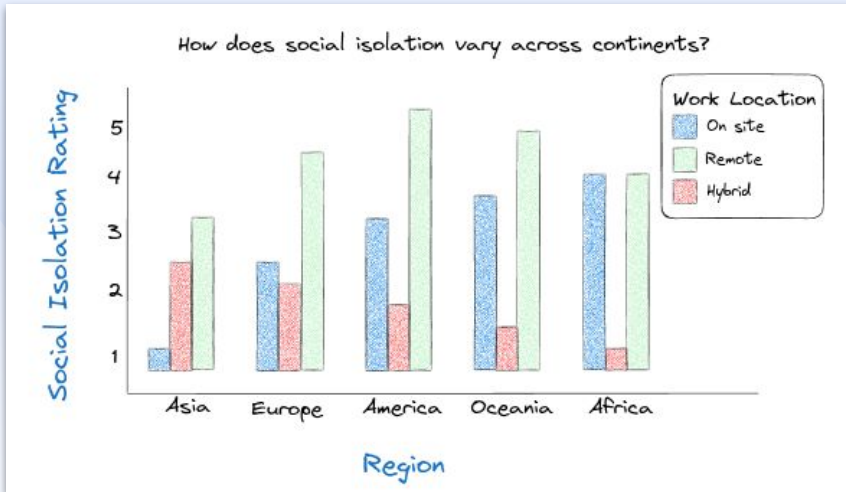


User Evaluation – Paper Prototype

Task example:

Task 2 - Grouped Bar chart

Compare America and Europe. In which continent do employees feel more isolated when working remotely?

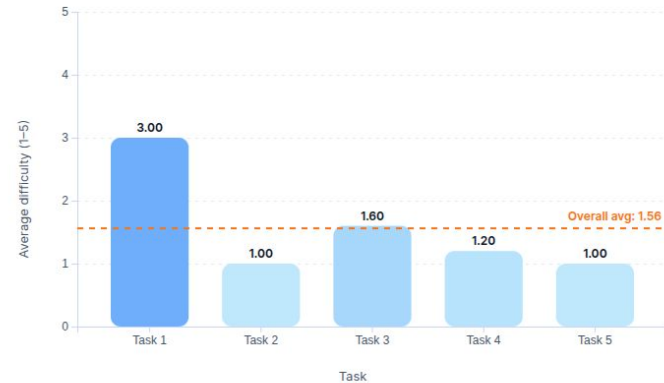


Task difficulty results

5 participants: 3 males and 2 female

Average task difficulty

How participants perceived the difficulty (1 = easy, 5 = hard)



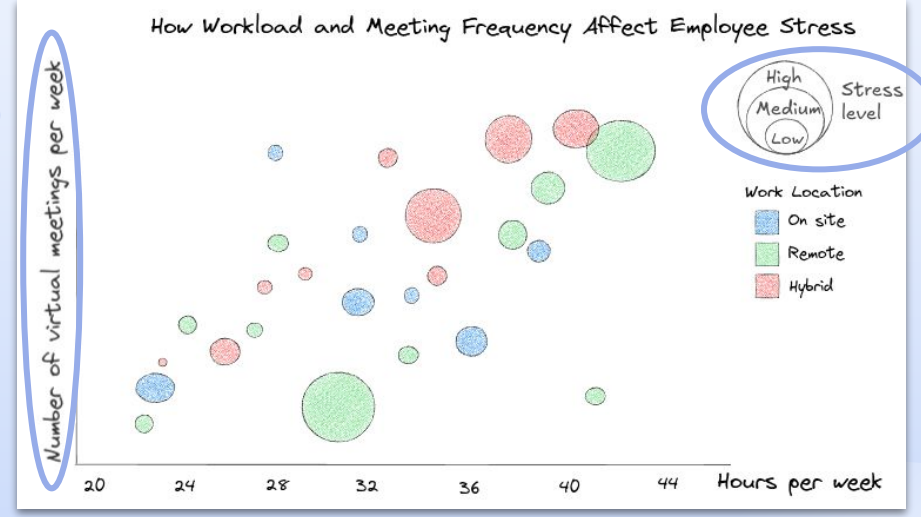
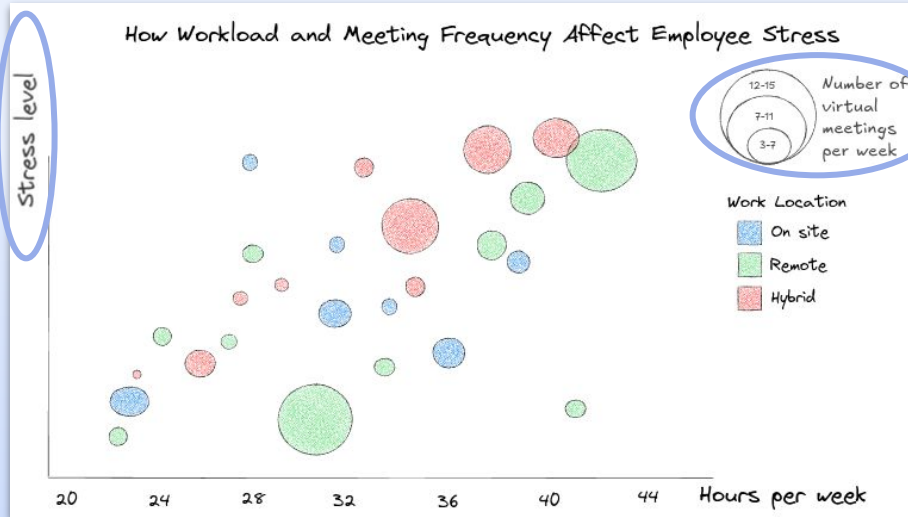
User Evaluation – Paper Prototype

“The bubble size would make more sense if it represented the stress level.”

Before

Bubble Chart

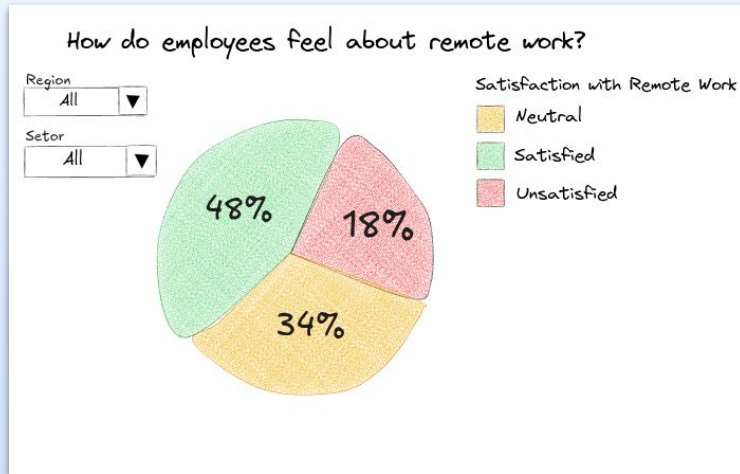
After



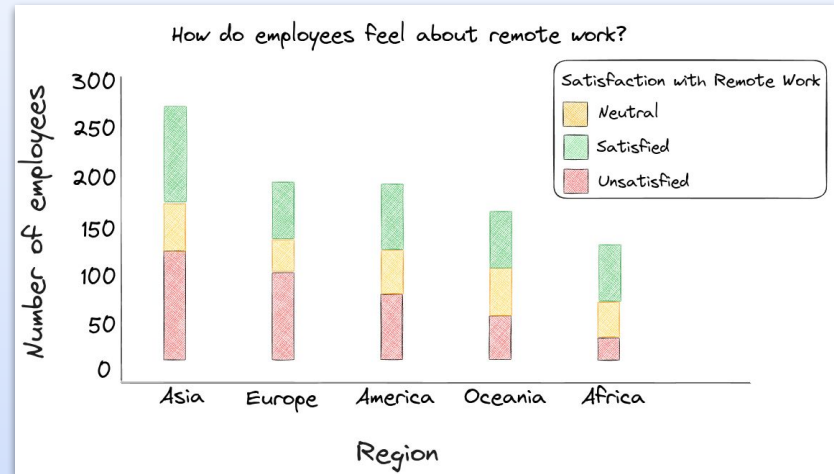
User Evaluation – Paper Prototype

“A stacked bar chart offers clearer regional comparison and works better with filtering and analysis than a pie chart.”

Before



After



Pie Chart

Stacked bar Chart

Employee percentage

Employee number

Live Demo!

<https://mind-work.vercel.app/>



Implementation Approach



Interactive models

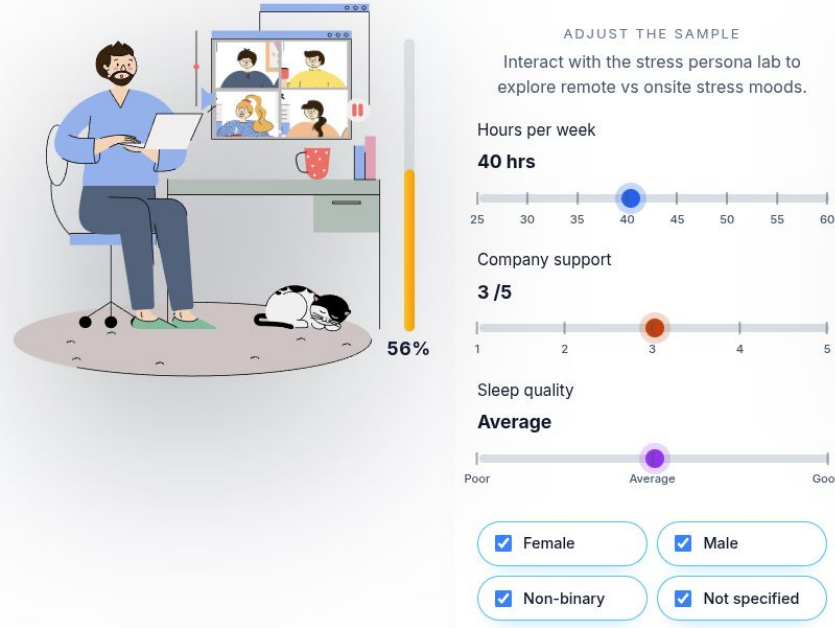
Overview dashboard

Workspace

Mental health

Documentation

Remote vs Onsite



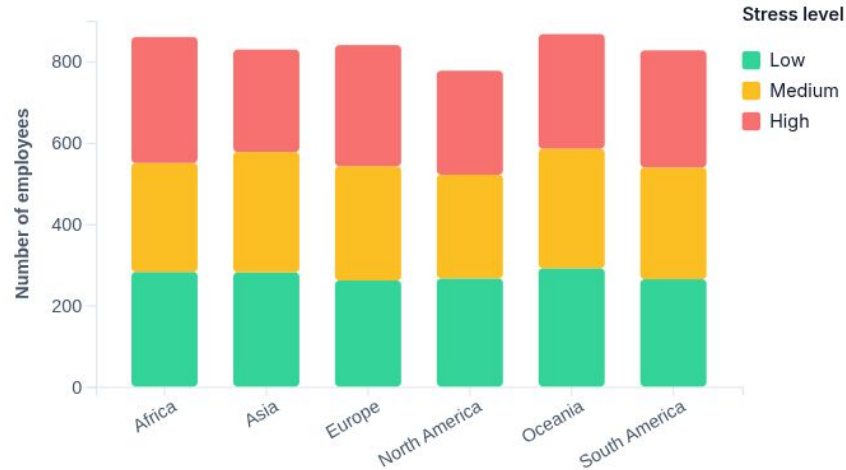
Implementation Approach

Schneiderman's viewing mantra:

1. Overview First
2. Zoom and Filter
3. Details on Demand

Stress levels by region

Start with the broad picture of how stress levels stack up in each region.



Implementation Approach

Stress levels by region

Start with the broad picture of how stress levels stack up in each region.

Use the dropdowns to filter the dataset

FILTER BY INDUSTRY

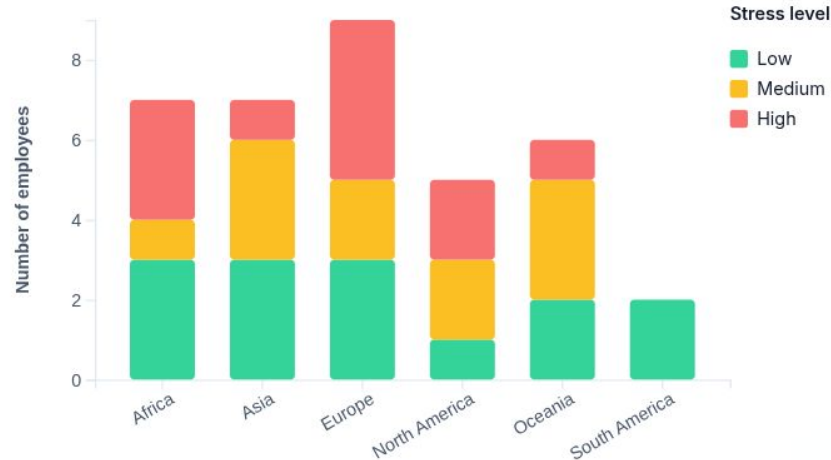
IT

FILTER BY JOB ROLE

Software Engineer

FILTER BY WORK LOCATION

Remote



Schneiderman's viewing mantra:

1. Overview First
2. Zoom and Filter
3. Details on Demand

Implementation Approach

Stress levels by region

Start with the broad picture of how stress levels stack up in each region.

Use the dropdowns to filter the dataset and click a bar to inspect that region in the details panel.

FILTER BY INDUSTRY

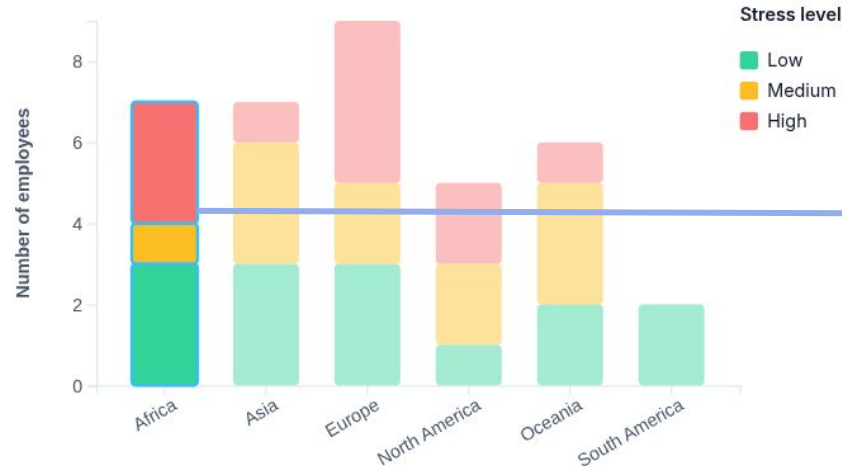
IT

FILTER BY JOB ROLE

Software Engineer

FILTER BY WORK LOCATION

Remote



Schneiderman's viewing mantra:

1. Overview First
2. Zoom and Filter
3. Details on Demand

Africa snapshot

EMPLOYEES

Metrics recalculated for the selected region.

7

PRODUCTIVITY CHANGE

Increase  3

No change  2

Decrease  2

REMOTE WORK SATISFACTION

Satisfied  4

Neutral  2

Unsatisfied  1

PHYSICAL ACTIVITY

Daily  2

Weekly  2

No activity  3

Implementation Approach



Interactive models

Overview dashboard

Workspace

Mental health

Documentation

Workspace: comparisons between different work arrangements (remote, hybrid, onsite).

Mental health: topics related to psychological support and work-life balance.

Documentation: support, explanations, and dataset variables.

Heuristic Evaluation

1 - Visibility of System Status

No problems found

2 - Match Between the System and the Real World

No problems found

3 - User Control and Freedom

Users can change or reset filters easily, but refreshing the page clears all selections, reducing their sense of control.

4 - Consistency and Standards

No problems found.

5 - Error Prevention

No problems found

6 - Recognition Rather than Recall

Inline examples beside filters would further minimize reliance on memory

7 - Flexibility and Efficiency of Use

No problems found

8 - Aesthetic and Minimalist Design

No problems found

9 - Help Users Recognize, Diagnose, and Recover from Errors

No problems found

10: Help and Documentation

No problems found

User Evaluation

5 Tasks and SUS Questionnaire

Task 1 - Determine the stress percentage

Using the Remote section, identify the percentage of employees who meet the following criteria...

Task 2 - Determine the Anxiety numbers

Using the Mental health conditions by region page (...)
Identify which region has the highest number of employees with Anxiety.

Task 3 – Explore isolation levels across regions

Identify which region has the highest average isolation rating in Hybrid.

Task 4 – Explore work-life balance for employees

Report the average work-life balance value at 15 years of experience for employees without mental health support.

Task 5 – Explore the role distribution in a specific sector and work location

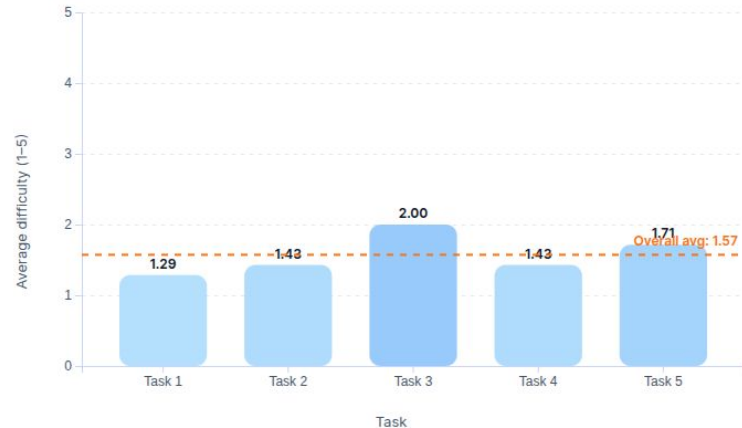
Find the number of Software Engineer people.

7 participants: 5 males and 2 female

Ages between 20 - 22

Average task difficulty

How participants perceived the difficulty (1 = easy, 5 = hard)



System Usability Scale (SUS)

Q1 ▾	Q2 ▾	Q3 ▾	Q4 ▾	Q5 ▾	Q6 ▾	Q7 ▾	Q8 ▾	Q9 ▾	Q10 ▾	SUS Score ▾
5	3	4	1	4	2	4	2	3	3	72,5
4	2	4	3	4	2	4	1	4	2	75
5	3	4	2	4	2	5	2	4	1	80
5	1	5	1	5	1	3	1	5	1	95
5	1	4	1	4	1	5	1	4	1	92,5
4	2	5	1	4	2	4	2	3	2	77,5
2	2	3	1	4	2	3	1	4	1	72,5
										77,5



User Evaluation – Feedback

Before

REMOTE WORK MENTAL HEALTH EXPLORER

Workspace

Workplace mix insights that combine bubble trends with isolation splits.

After

REMOTE WORK MENTAL HEALTH EXPLORER

Workspace

Workplace mix insights that combine bubble trends with isolation splits.

Hover over bubbles and bars for detailed information and ratings.



User Evaluation – Feedback

Before

REMOTE WORK MENTAL HEALTH EXPLORER

Overview dashboard

Global stress distribution and mental-health indicators.

Switch to dark mode Switch to Portuguese

Stress levels by region

Start with the broad picture of how stress levels stack up in each region.

Stress level Mental health condition

After

REMOTE WORK MENTAL HEALTH EXPLORER

Overview dashboard

Global stress distribution and mental-health indicators.

Toggle between stress level and mental health condition views.

Switch to dark mode Switch to Portuguese

Mental health conditions by region

Start with the broad picture of how stress levels stack up in each region.

Stress level Mental health condition

Limitations

Stress and burnout scores rely on self-report, which can be influenced by recall bias or mood at submission

Variables like team size, economic conditions are not controlled

The values chosen for variables such as stress, productivity, and sleep quality were not ideal

The results of a potential query may not yield meaningful or reliable outcomes.

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